Schur-Rank Multiplication Tensors and Structural Rigidity of Determinant Orbit Closures

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Abstract

We introduce the Schur-Rank Multiplication Tensor (SRMT) of a graded G-algebra $A = \mathbb{C}[\mathcal{O}]$,

$$\mathrm{sr}_A^{d_1,d_2}(\lambda,\mu;\nu) := \mathrm{rank}\big(m_\nu\colon (A_{d_1})_\lambda \otimes (A_{d_2})_\mu \to A_{d_1+d_2,\nu}\big),$$

where m_ν is the ν-isotypic component of the multiplication map. We show that the SRMT carries strictly more information than the collection of isotypic multiplicities $\{c_\lambda(A,d)\}$ (Proposition 3.1), and is therefore not obstructed by the Bürgisser–Ikenmeyer–Panova theorem [1].

Let $G = \mathrm{GL}_{m^2}$, $H = \mathrm{GL}_m \times \mathrm{GL}_m$, and let $O_{\det_m}$, O_{perm_n} be the G-orbit closures of the determinant and permanent in $W = \mathrm{Sym}_m(\mathbb{C}^{m^2*})$. The unconditional results proved in this paper are the following.

(i) **Ray-Multiplicity-1 for the determinant:** the H-weight $k\lambda_{\det}$ occurs with multiplicity exactly 1 in $\mathbb{C}[O_{\det_m}]_k$ for every $k \geq 0$, where $\lambda_{\det} = (\omega_m;\omega_m)$ is the H-weight of $\det_m$ (Theorem 4.4). Every SRMT value at weight triples along the ray $(a\lambda_{\det}, b\lambda_{\det}; (a+b)\lambda_{\det})$ consequently lies in $\{0,1\}$. Full multiplicity-freeness of $\mathbb{C}[O_{\det_m}]$ as an H-module is expected but not established here; see Conjecture 4.6.

(ii) **Bilinear Rank-1 Rigidity:** for all $a, b \geq 0$ with $a + b \geq 1$,

$$\mathrm{sr}^{a,b,H}_{O_{\det_m}}(a\lambda_{\det}, b\lambda_{\det}; (a+b)\lambda_{\det}) = 1.$$

The H-isotypic sub-algebra $\bigoplus_{k\geq 0}(\mathbb{C}[O_{\det_m}]_k)_{k\lambda_{\det}}$ is isomorphic, as a graded ring, to $\mathbb{C}[t]$ with $\deg t = 1$.

(iii) For $n \geq 2$, the diagonal torus $T \cong (\mathbb{C}^*)^{2n-1}$ (the identity component of the stabiliser of perm_n in GL_{n^2}) acts by character on perm_n. Note that $\det_m$ and perm_n share the *same* torus weight $(\omega_m;\omega_m)$ as T-characters; they differ at the H-module level, where perm_n lies in the Schur component $S_{(n)} \otimes S_{(n)}$ of highest weight $\mu_{\mathrm{perm}} = (n\omega_1;\, n\omega_1)$. The fundamental obstruction to Rank-1 Rigidity lies in the high-dimensionality of this H-module: $\dim L^H(\mu_{\mathrm{perm}}) = \binom{2n-1}{n}^2 \gg 1$ for $n \geq 2$, in stark contrast to $\dim L^H(\lambda_{\det}) = 1$. This non-uniqueness of the μ_{perm}-isotypic weight space is the fundamental obstruction preventing any Rank-1 Rigidity argument from first principles for the permanent.

(iv) For $n \geq 2$, the orbit $G \cdot \mathrm{perm}_n$ is not spherical for $G = \mathrm{GL}_{n^2}$ (Proposition 8.1), so Brion's multiplicity-free criterion fails. Whether $\mathbb{C}[O_{\mathrm{perm}_n}]$ has some isotypic multiplicity ≥ 2 is the content of Open Problem (S1), expected to hold.

(v) SRMT values are monotone under orbit-closure inclusion at a fixed degree pair: for $O_1 \subseteq O_2$ and fixed $d_1, d_2 \geq 0$, $\mathrm{sr}_{O_1}^{d_1,d_2}(\lambda, \mu; \nu) \leq \mathrm{sr}_{O_2}^{d_1,d_2}(\lambda, \mu; \nu)$ for all (λ, μ, ν).

Together, (i)–(v) establish a provable structural asymmetry between the two coordinate rings, one invisible to any module-level analysis. The resulting computational separation programme is formulated as Open Problems (S1)–(S4) (Section 9), each reducible to finite-dimensional linear algebra over $\mathbb{C}$ executable in Macaulay2. The three primary problems are logically independent: (S1) and (S2) concern the permanent's SRMT lower bound; (S3) concerns the determinant's SRMT upper bound at the same weight triple; (S4) is a structural question about orbit-level multiplicity–SRMT independence, not required for the separation but relevant to the general programme.

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1 Introduction

Background

Valiant's conjecture [6] asserts that perm_n is not a polynomial-size linear projection of $\det_m$. In characteristic zero this is equivalent to the strict non-containment of G-orbit closures

$$O_{\mathrm{perm}_n} \not\subseteq O_{\det_m} \subset W = \mathrm{Sym}_m(\mathbb{C}^{m^2 *}).$$

The Geometric Complexity Theory (GCT) programme of Mulmuley–Sohoni [4, 5] proposes to witness this non-containment via *occurrence obstructions*: dominant weights λ such that $c_\lambda(\mathbb{C}[O_{\mathrm{perm}_n}]) > 0$ but $c_\lambda(\mathbb{C}[O_{\det_m}]) = 0$.

The 2019 theorem of Bürgisser, Ikenmeyer, and Panova [1] (hereafter BIP) closes this approach: for m polynomial in n, the multiplicity functions $\lambda \mapsto c_\lambda(\mathbb{C}[O_{\det_m}])$ and $\lambda \mapsto c_\lambda(\mathbb{C}[O_{\mathrm{perm}_n}])$ are asymptotically indistinguishable. No occurrence obstruction exists.

The BIP theorem is, however, a theorem about *modules*, not *algebras*. It says nothing about the multiplicative structure of the coordinate rings, the ranks of their multiplication maps, or their syzygy modules. The present paper shows that this discarded information carries a provable structural distinction between $\mathbb{C}[O_{\det_m}]$ and $\mathbb{C}[O_{\mathrm{perm}_n}]$.

The SRMT

For a connected graded G-algebra $A = \bigoplus_{d\geq 0} A_d$, the *Schur-Rank Multiplication Tensor* at degree pair (d_1, d_2) is

$$\mathrm{sr}_A^{d_1,d_2}(\lambda, \mu; \nu) := \mathrm{rank}\big(m_\nu \colon (A_{d_1})_\lambda \otimes (A_{d_2})_\mu \to A_{d_1+d_2,\nu}\big),$$

where m_ν is the ν-isotypic component of the multiplication map. When the context is clear we drop the superscript and write $\mathrm{sr}_A(\lambda, \mu; \nu)$, with the understanding that d_1, d_2 are specified (often as the smallest degrees at which λ, μ first appear in A; see Definition 2.3 and Remark 2.4).

Proposition 3.1 shows that the SRMT is strictly finer than the full multiplicity table $\{c_\lambda(A, d)\}$: there exist graded G-algebras with identical isotypic multiplicities but distinct SRMT values. The SRMT is therefore not closed off by the BIP theorem.

Logical structure of the separation argument

We emphasize at the outset a point that is elaborated in Remarks 3.4 and 3.2: the module–algebra gap (Proposition 3.1) plays a specific and limited role.

- **Role of Proposition 3.1.** It establishes that the map $A \mapsto \{c_\lambda(A, d)\}$ is *not injective* on the class of all connected graded G-algebras; the SRMT is supported on its kernel. This shows that BIP does not, in principle, obstruct SRMT-based separation arguments for any graded G-algebra.

- **Role of Sections 5–8.** The unconditional results of these sections compute SRMT values for the specific orbit-closure coordinate rings $\mathbb{C}[O_{\mathrm{det}_m}]$ and $\mathbb{C}[O_{\mathrm{perm}_n}]$ *directly*, without any appeal to the abstract gap of Proposition 3.1. In particular, Bilinear Rank-1 Rigidity (Theorem 5.4) establishes $\mathrm{sr}^H_{O_{\mathrm{det}_m}}(a\lambda_{\mathrm{det}}, b\lambda_{\mathrm{det}}; (a+b)\lambda_{\mathrm{det}}) = 1$ by an explicit character computation, not by transfer from the abstract setting.

- **Role of Open Problems (S1)–(S3).** The actual GCT separation (if established) proceeds via SRMT Monotonicity (Theorem 7.1), using directly computed SRMT values that diverge between $\mathbb{C}[O_{\mathrm{perm}_n}]$ and $\mathbb{C}[O_{\mathrm{det}_m}]$. No assumption is made that the geometric structure of orbit closures prevents multiplicities from determining SRMT values; this potential rigidity question is isolated as Open Problem (S4) and does not affect the separation argument via (S1)–(S3).

Main results and status discipline

Throughout the paper, we maintain a strict distinction between what is proved and what remains open.

- **Proved unconditionally:** all theorems, lemmas, propositions, corollaries, and examples in Sections 2–7 and Appendix C.

- **Open problems:** the four computational conditions (S1)–(S4) of Section 9.

- **Conjectural (clearly labelled):** full multiplicity-freeness of $\mathbb{C}[O_{\mathrm{det}_m}]$ as an H-module (Conjecture 4.6), the Koszul structure, and the Newton–Okounkov volume reformulation, discussed in Section 10.

The core unconditional contribution is Bilinear Rank-1 Rigidity (Theorem 5.4): for every $a, b \geq 0$ with $a + b \geq 1$,

$$\mathrm{sr}^H_{O_{\mathrm{det}_m}}(a\lambda_{\mathrm{det}}, b\lambda_{\mathrm{det}}; (a+b)\lambda_{\mathrm{det}}) = 1.$$

This follows from two independently provable facts: the character identity $\det_m(AXB^T) = \det(A)\det(B)\det_m(X)$ and the one-dimensionality of $L^H(k\lambda_{\mathrm{det}})$. No invocation of sphericity or Brion's theorem is required.

Corrections to earlier drafts

We draw attention to three points where the present version corrects errors in earlier drafts.

- *False sphericity argument.* An earlier version invoked Brion's theorem to conclude that $\mathbb{C}[O_{\mathrm{det}_m}]$ is multiplicity-free as an H-module. This was incorrect: O_{det_m} is not spherical for H (Remark 4.3). The corrected paper replaces this by Theorem 4.4 (Ray-Multiplicity-1), which suffices for all subsequent results. Full multiplicity-freeness is demoted to Conjecture 4.6.

- *Wrong contrapositive of Brion.* An earlier version incorrectly concluded "not spherical $\Rightarrow$ not multiplicity-free" for the permanent orbit. The correct implication runs the other way. Proposition 8.1 now states only the non-sphericity; non-multiplicity-freeness is relegated to Open Problem (S1) with a careful explanation (Remark 8.2).

- *Erroneous LR coefficient.* An earlier version claimed $N^{2\mu_{\mathrm{perm}}}_{\mu_{\mathrm{perm}},\mu_{\mathrm{perm}}} > 1$ for $m \geq 2$. The correct value is 1 by the Pieri rule; see Remark 2.12. The SRMT flexibility argument for the permanent is consequently reformulated: it rests on $c_{\mu_{\mathrm{perm}}}(\mathbb{C}[O_{\mathrm{perm}_n}], d) \geq 2$ (Open Problem (S1)), not on the LR coefficient.

Organisation

Section 2 establishes notation, the formal definition of the SRMT, and the Cauchy decomposition. Section 3 proves the module–algebra gap. Section 4 proves Ray-Multiplicity-1. Section 5 contains Bilinear Rank-1 Rigidity. Section 6 establishes the SRMT–syzygy correspondence. Section 7 proves SRMT Monotonicity. Section 8 treats the permanent's SRMT. Section 9 presents the computational separation programme. Section 10 collects conjectural extensions.

2 Definitions and Conventions

Throughout, $\mathbb{C}$ denotes the field of complex numbers; all algebraic groups and varieties are defined over $\mathbb{C}$. Fix integers $m, n \geq 1$ (with $m \geq n$ when comparing the two orbit closures). Set $G = \mathrm{GL}_{m^2}(\mathbb{C})$ and $W = \mathrm{Sym}_m(\mathbb{C}^{m^2*})$ (degree-m homogeneous polynomials in m^2 variables). The group G acts on W by linear substitution: $(g \cdot f)(x) = f(g^T x)$.

2.1 Groups and orbit closures

Definition 2.1 (Orbit closures)**.**

$$O_{\det_m} := \overline{G \cdot \det_m} \subset W, \qquad O_{\mathrm{perm}_n} := \overline{G \cdot \mathrm{perm}_n} \subset W,$$

where the closures are taken in the Zariski topology. We maintain the strict distinction between the open G-orbit $G \cdot \det_m$ and its Zariski closure $O_{\det_m}$. All results of this paper concern the closures.

Set $H = \mathrm{GL}_m(\mathbb{C}) \times \mathrm{GL}_m(\mathbb{C})$, embedded in G via $(A, B) \cdot X = AXB^T$ for $(A, B) \in H$ and $X \in M_{m\times m}(\mathbb{C})$. The variety $O_{\det_m}$ is the closure of the G-orbit of $\det_m$; the group H acts on it by restriction of the G-action.

Remark 2.2 (Roles of G and H; automorphism groups)**.** *The orbit closures of Definition 2.1 are always closures of $G = \mathrm{GL}_{m^2}$-orbits. The H-orbit of $\det_m$ in W is the one-dimensional subspace $\mathbb{C}^* \cdot \det_m$ (since the character identity shows H acts on $\det_m$ by the scalar $\det(A)\det(B)$). All GCT-relevant geometry lives at the G-level; the H-module structure of $\mathbb{C}[O_{\det_m}]$ is an additional algebraic tool, not a replacement for the G-action.*

The full automorphism group of $\det_m$ as a polynomial (in the sense of linear transformations of $\mathbb{C}^{m^2}$ preserving $\det_m$) is

$$\mathrm{Aut}(\det_m) \cong (\mathrm{GL}_m \times \mathrm{GL}_m)/\mu_m \rtimes \mathbb{Z}/2\mathbb{Z},$$

where $\mu_m = \{(\zeta I_m, \zeta^{-1} I_m) : \zeta^m = 1\}$ is the diagonal scalar subgroup (acting trivially on $\det_m$) and the $\mathbb{Z}/2\mathbb{Z}$ factor corresponds to transposition $X \mapsto X^T$. The subgroup $H = \mathrm{GL}_m \times \mathrm{GL}_m$ contains the diagonal torus $(\mathbb{C}^*)^{2m}$, the unipotent radical of each Borel, and the Weyl group $S_m \times S_m$ (permutation matrices). The identity component of $\mathrm{Aut}(\det_m)$ is $(\mathrm{GL}_m \times \mathrm{GL}_m)/\mu_m$, of dimension $2m^2 - 1$.

2.2 Representations and multiplicities

Let $\Lambda^+ = \Lambda^+(G)$ denote the dominant integral weights of G. For a rational G-module V and $\lambda \in \Lambda^+$:

$$c_\lambda(V) := \dim \mathrm{Hom}_G(L(\lambda), V) \quad \text{(multiplicity of the irreducible } L(\lambda) \text{ in } V\text{)}.$$

For a graded G-algebra $A = \bigoplus_{d\geq 0} A_d$: $c_\lambda(A, d) := c_\lambda(A_d)$.

The Littlewood–Richardson coefficient $N^\nu_{\lambda\mu} := \dim \mathrm{Hom}_G(L(\nu), L(\lambda) \otimes L(\mu))$ controls how irreducibles tensor.

The *weight semigroup* of an orbit closure $\mathcal{O}$ is $\mathcal{X}(\mathcal{O}) := \{\lambda \in \Lambda^+ \mid c_\lambda(\mathbb{C}[\mathcal{O}]) > 0\}$; for H-weights we write $\mathcal{X}_H(\mathcal{O})$.

2.3 Formal definition of the SRMT

Definition 2.3 (SRMT and SRMT defect)**.** *Let $A = \bigoplus_{d\geq 0} A_d$ be a connected graded G-algebra and let $d_1, d_2 \geq 0$ be integers. For $\lambda, \mu, \nu \in \Lambda^+$ define the* degree-(d_1, d_2) SRMT*:*

$$\mathrm{sr}_A^{d_1,d_2}(\lambda, \mu; \nu) := \mathrm{rank}\big(m_\nu \colon (A_{d_1})_\lambda \otimes (A_{d_2})_\mu \to A_{d_1+d_2,\nu}\big),$$

$$\delta_A^{d_1,d_2}(\lambda, \mu; \nu) := \min\big(N^\nu_{\lambda\mu}\, c_\lambda(A, d_1)\, c_\mu(A, d_2),\ c_\nu(A, d_1 + d_2)\big) - \mathrm{sr}_A^{d_1,d_2}(\lambda, \mu; \nu) \geq 0,$$

where m_ν is the ν-isotypic component of the multiplication map $m_{d_1,d_2} \colon A_{d_1} \otimes A_{d_2} \to A_{d_1+d_2}$. We call $\delta_A^{d_1,d_2}(\lambda, \mu; \nu)$ the SRMT defect*.*

When d_1 (resp. d_2) is the smallest degree in which $c_\lambda(A, d_1) > 0$ (resp. $c_\mu(A, d_2) > 0$), we drop the superscript and write $\mathrm{sr}_A(\lambda, \mu; \nu)$, $\delta_A(\lambda, \mu; \nu)$.

Write $\mathrm{SRMT}(A) := \{\mathrm{sr}_A^{d_1,d_2}(\lambda, \mu; \nu)\}$ for the full collection of rank values over all admissible $(d_1, d_2, \lambda, \mu, \nu)$.

Remark 2.4 (Notation for degree pairs in cross-algebra comparisons)**.** *The SRMT without superscripts $\mathrm{sr}_A(\lambda, \mu; \nu)$ uses the minimal degrees at which λ and μ first appear in A. These minimal degrees are intrinsic to A and may differ between two algebras A and B:*

$$d_1^A := \min\{d : c_\lambda(A, d) > 0\}, \qquad d_1^B := \min\{d : c_\lambda(B, d) > 0\}.$$

When $O_1 \subseteq O_2$ and $c_\lambda(\mathbb{C}[O_1], d) > 0$ first at degree $d_1^{(1)}$ while $c_\lambda(\mathbb{C}[O_2], d) > 0$ first at degree $d_1^{(2)}$, one typically has $d_1^{(1)} \geq d_1^{(2)}$ (the larger orbit contains the class at smaller or equal degree). A comparison of $\mathrm{sr}_{\mathbb{C}[O_1]}^{d_1^{(1)},d_2^{(1)}}$ with $\mathrm{sr}_{\mathbb{C}[O_2]}^{d_1^{(2)},d_2^{(2)}}$ involves different degree pairs *and falls outside the monotonicity framework.*

Convention for cross-algebra comparisons. *Whenever SRMT values for two different algebras or orbit closures are compared (as in the separation programme of Section 9), the degree pair (d_1, d_2) is fixed to be the* same *explicit integer pair for both algebras:*

$$\mathrm{sr}_{O_{\mathrm{perm}_3}}^{d_0,d_0}(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}}) \quad \textit{vs.} \quad \mathrm{sr}_{O_{\det_3}}^{d_0,d_0}(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}}),$$

where d_0 is the degree determined by Open Problem (S1). In this setting the superscript is never dropped. All degree pairs in the BIP regime (Remark 2.5) are also explicitly specified.

Remark 2.5 (The SRMT and the BIP Barrier)**.** *The SRMT is specifically designed to operate in the regime where the BIP theorem provides no information. Concretely, BIP establishes that for m polynomial in n, the isotypic multiplicities satisfy $c_\mu(\mathbb{C}[O_{\mathrm{det}_m}], d) = c_\mu(\mathbb{C}[O_{\mathrm{perm}_n}], d)$ for all (μ, d) in a large asymptotic range (Scenario B). In Scenario B, no module-level obstruction can distinguish the two orbit closures.*

The SRMT is built precisely to go beyond Scenario B: even when $c_\mu(perm) = c_\mu(det)$, the ranks of the isotypic multiplication maps $m_\nu\colon (A_{d_1})_\lambda \otimes (A_{d_2})_\mu \to A_{d_1+d_2,\nu}$ can differ between the two algebras. This rank information is invisible to any module-theoretic analysis and is therefore not closed off by BIP.

Caveat. *Whether the SRMT is actually independent of multiplicities* for orbit-closure coordinate algebras specifically *(not just for abstract graded algebras) is a separate question, isolated as Open Problem (S4). The separation argument of Section 9 does not require (S4): it uses directly computed SRMT values that differ between $\mathbb{C}[O_{\mathrm{det}_m}]$ and $\mathbb{C}[O_{\mathrm{perm}_n}]$.*

Remark 2.6 (Why degree pairs matter for monotonicity)**.** *The definition of SRMT at minimal degrees is natural for intrinsic properties of a single algebra. For comparisons between two algebras A_1, A_2 linked by a surjection—in particular for the Monotonicity Theorem (Theorem 7.1)—one must fix the degree pair (d_1, d_2) to be the same for both algebras. This is discussed carefully in Appendix C.*

Definition 2.7 (SRMT Monotonicity)**.** *If $O_1 \subseteq O_2$ are G-orbit closures and $d_1, d_2 \geq 0$ are fixed degrees, the Zariski-closed inclusion induces a surjection of graded G-algebras $\mathbb{C}[O_2] \twoheadrightarrow \mathbb{C}[O_1]$, which implies (see Theorem 7.1 and Appendix C)*

$$\mathrm{sr}_{O_1}^{d_1,d_2}(\lambda, \mu; \nu) \leq \mathrm{sr}_{O_2}^{d_1,d_2}(\lambda, \mu; \nu) \quad \textit{for all triples } (\lambda, \mu, \nu).$$

In particular, any strict inequality $\mathrm{sr}_{O_1}^{d_1,d_2}(\lambda, \mu; \nu) > \mathrm{sr}_{O_2}^{d_1,d_2}(\lambda, \mu; \nu)$ at a single triple and degree pair certifies $O_1 \not\subseteq O_2$.

2.4 Cauchy decomposition and the H-module structure of W

Proposition 2.8 (Cauchy decomposition; H-weights of det_m and perm_n)**.** *Let $U = V = \mathbb{C}^m$, so that $\mathbb{C}^{m^2} \cong U \otimes V$. The following hold as $H = \mathrm{GL}(U) \times \mathrm{GL}(V)$-module isomorphisms.*

(i) (Cauchy identity.) *There is a canonical H-equivariant decomposition*

$$W = \mathrm{Sym}_m(U^* \otimes V^*) \cong \bigoplus_{\lambda \vdash m} S_\lambda(U^*) \otimes S_\lambda(V^*),$$

where the sum runs over all partitions λ of m and S_λ denotes the Schur functor associated with λ. In particular, W is a multiplicity-free H-module.

(ii) (Determinant.) *The polynomial det_m is, up to a nonzero scalar, the unique vector of H-weight λ_{det} in the component $\lambda = (1^m)$:*

$$S_{(1^m)}(U^*) \otimes S_{(1^m)}(V^*) = \bigwedge^m U^* \otimes \bigwedge^m V^*.$$

This component has dimension 1; its H-highest weight is $\lambda_{\mathrm{det}} = (\omega_m; \omega_m)$, where ω_m is the m-th fundamental weight of GL_m (the determinant character).

(iii) (Permanent.) *The polynomial* perm_n *is, up to a nonzero scalar, the unique vector of* H*-weight* μ_{perm} *in the component* $\lambda = (m)$*:*

$$S_{(m)}(U^*) \otimes S_{(m)}(V^*) = \mathrm{Sym}^m(U^*) \otimes \mathrm{Sym}^m(V^*).$$

Its H*-highest weight is* $\mu_{\mathrm{perm}} = (m\omega_1; m\omega_1)$*, where* ω_1 *is the first fundamental weight of* GL_m*.*

(iv) (Global G-structure.) *As a* $G = \mathrm{GL}_{m^2}$*-module,* W *is irreducible of highest weight* $m\omega_1^{\mathrm{GL}_{m^2}}$*. Consequently,* $\det_m$ *and* perm_n *belong to the same* GL_{m^2}*-module; the distinction between them is an* H*-module phenomenon.*

Remark 2.9 (Torus weights versus Schur module labels: a critical distinction)**.** *Proposition 2.8 introduces two notions that must not be confused.*

Torus weight (character). *Both* $\det_m$ *and* perm_n *are eigenvectors for the diagonal maximal torus* $T_H = \{(\mathrm{diag}(t_1, \ldots, t_m), \mathrm{diag}(s_1, \ldots, s_m))\} \subset H$ *acting by the character* $(t_1, \ldots, t_m; s_1, \ldots, s_m) \mapsto \prod_i t_i \cdot \prod_j s_j$*. As torus characters,* $\det_m$ *and* perm_n *have* identical *torus weight* $(\omega_m; \omega_m) = (1, \ldots, 1; 1, \ldots, 1)$ *in additive notation. This is because every monomial of* $\det_m$ *and every monomial of* perm_n *involves each row variable and each column variable exactly once.*

H-highest weight (Schur module label). *The two polynomials live in* different H*-irreducible components of* W*:*

$$\det_m \in S_{(1^m)}(U^*) \otimes S_{(1^m)}(V^*), \quad H\text{-highest weight } \lambda_{\det} = (\omega_m; \omega_m) \text{ (one-dimensional)};$$

$$\mathrm{perm}_n \in S_{(m)}(U^*) \otimes S_{(m)}(V^*), \quad H\text{-highest weight } \mu_{\mathrm{perm}} = (m\omega_1; m\omega_1) \text{ (dimension } \tbinom{2m-1}{m}^2\text{)}.$$

Key practical consequence. *The notation* $\mu_{\mathrm{perm}} = (m\omega_1; m\omega_1)$ *refers to the* H-highest weight of the Schur module *in which* perm_n *resides,* not *to the torus character of* perm_n*. The torus character of* perm_n *is* $(\omega_m; \omega_m)$*, identical to that of* $\det_m$*. All multiplicity computations and SRMT analysis in this paper operate at the level of* H*-highest weights (equivalently,* H*-isotypic components),* not *at the level of torus characters. Concretely: the weight* μ_{perm} *indexes the Schur component* $\mathrm{Sym}^m U^* \otimes \mathrm{Sym}^m V^*$*, whereas* $\lambda_{\det}$ *indexes the Schur component* $\bigwedge^m U^* \otimes \bigwedge^m V^*$*. These are orthogonal summands of* W *under the Cauchy decomposition (Proposition 2.8(i)), so* $c_{k\lambda_{\det}}(W, 1) = 1$ *while* $c_{k\lambda_{\det}}(W, 1)$ *for* $k \geq 2$ *requires passing to* $\mathrm{Sym}^k(W)$*. The distinction is essential: two polynomials with the same torus weight can lie in completely different* H*-isotypic components and exhibit radically different SRMT behaviour.*

Remark 2.10 (Notation convention for elements of the coordinate ring)**.** *We systematically distinguish two objects:*

- $\det_m \in W$*: the determinant polynomial, a point in the vector space* W*. Under* H*, it transforms as* $((A, B) \cdot \det_m)(X) = \det_m(A^T X B) = \det(A)\det(B)\det_m(X)$*, so it is an* H*-eigenvector of weight* $\lambda_{\det} = (\omega_m; \omega_m)$*.*

- $\widehat{\det}_m \in A_1 = (\mathbb{C}[O_{\det_m}])_1$*: the restriction to* $O_{\det_m}$ *of the unique (up to scalar)* H*-eigenvector of weight* $\lambda_{\det}$ *in* $W^* \cong \mathbb{C}[W]_1$*.*

More precisely: the element $\widehat{\det}_m$ is the image of the linear functional $\ell_{\det_m} \in W^*$ defined by $\ell_{\det_m}(f) = \langle f, \det_m \rangle$ under the restriction map $\mathbb{C}[W]_1 \twoheadrightarrow A_1 = \mathbb{C}[O_{\det_m}]_1$; its H-weight as an element of A_1 is $\lambda_{\det}$. Henceforth, when we write $\det_m \in A_k$ or $(\det_m)^k \in A_k$, we mean $\widehat{\det}_m^{\,k}$—the k-th power of the corresponding element of A_1. This is an element of the coordinate ring; the H-weight of $(\det_m)^k \in A_k$ is $k\lambda_{\det}$.

Remark 2.11. *For $m \geq 2$ the weights $\lambda_{\det} = (\omega_m; \omega_m)$ and $\mu_{\mathrm{perm}} = (m\omega_1; m\omega_1)$ are distinct elements of $\Lambda^+(H)$. Indeed, ω_1 and ω_m are the extreme fundamental weights of GL_m; they coincide only for $m = 1$, and for $m \geq 2$ the partitions (1^m) and (m) are distinct. In particular $\mu_{\mathrm{perm}} \neq k\lambda_{\det}$ for any positive integer k. All weight-based distinctions in this paper operate at the H-weight level, not the GL_{m^2}-weight level.*

Remark 2.12 (G-weights versus H-weights; Littlewood–Richardson for $H = \mathrm{GL}_m \times \mathrm{GL}_m$)**.** *Throughout this paper we work with two distinct notions of weight.*

G-weights *($G = \mathrm{GL}_{m^2}$). An irreducible G-module $L(\lambda)$ is labelled by a dominant weight $\lambda \in \Lambda^+(G)$, a non-increasing sequence of integers $\lambda_1 \geq \cdots \geq \lambda_{m^2}$.*

H-weights *($H = \mathrm{GL}_m \times \mathrm{GL}_m$). An irreducible H-module is of the form $L^H(\alpha; \beta) = S_\alpha(\mathbb{C}^m) \otimes S_\beta(\mathbb{C}^m)$, where $\alpha, \beta \in \Lambda^+(\mathrm{GL}_m)$. By Schur's lemma, all computations of isotypic multiplicities and SRMT values in Sections 4–8 are performed at the H-weight level.*

Littlewood–Richardson coefficients for H. *For the product group $H = \mathrm{GL}_m \times \mathrm{GL}_m$, the tensor product rule factorises:*

$$L^H(\alpha; \beta) \otimes L^H(\alpha'; \beta') \cong \bigoplus_{\gamma, \delta} \left(N^{\gamma}_{\alpha\alpha'} N^{\delta}_{\beta\beta'} \right) L^H(\gamma; \delta),$$

where $N^{\gamma}_{\alpha\alpha'}$ and $N^{\delta}_{\beta\beta'}$ are the classical GL_m Littlewood–Richardson coefficients. In particular, the H-LR coefficient factorises as a product of two GL_m coefficients.

For weights of the form $(\lambda; \lambda)$ (as for $k\lambda_{\det} = (k\omega_m; k\omega_m)$), this gives:

$$N^{(a+b)\lambda_{\det}}_{a\lambda_{\det},\, b\lambda_{\det}} = \left(N^{(a+b)\omega_m}_{a\omega_m,\, b\omega_m} \right)^2 = 1^2 = 1,$$

since ω_m is a determinant character of GL_m and the tensor product of determinant characters is again a determinant character (classical).

LR coefficient for the permanent. *For $\mu_{\mathrm{perm}} = (m\omega_1; m\omega_1)$, the factorised H-LR coefficient is*

$$N^{2\mu_{\mathrm{perm}}}_{\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}} = \left(N^{2m\omega_1}_{m\omega_1,\, m\omega_1} \right)^2. \tag{1}$$

*By the **Pieri rule**, the product $S_{(m)} \otimes S_{(m)}$ decomposes over partitions obtainable from (m) by adding m boxes with no two in the same column. The only single-row partition of size $2m$ obtainable in this way is $(2m)$ itself, occurring with multiplicity 1. Thus $N^{2m\omega_1}_{m\omega_1,\, m\omega_1} = 1$, and*

$$\boxed{N^{2\mu_{\mathrm{perm}}}_{\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}} = 1.}$$

Warning (corrected from an earlier version). *A previous draft claimed that this coefficient "generally exceeds 1 for $m \geq 2$". This was an error. The correct value is 1 by the Pieri rule. The flexibility argument for the SRMT of $\mathbb{C}[O_{\mathrm{perm}_n}]$ does* not *rest on the LR coefficient exceeding 1; rather, it rests on the possibility that $c_{\mu_{\mathrm{perm}}}(\mathbb{C}[O_{\mathrm{perm}_n}], d) \geq 2$ for some degree d (see Open Problem (S1) and Remarks 8.3 and 5.9 below).*

3 The Module–Algebra Gap

We prove that the SRMT is not determined by the isotypic multiplicities $\{c_\lambda(A,d)\}$.

Proposition 3.1 (Module data does not determine SRMT for abstract graded algebras)**.** *There exist pairs of connected graded G-algebras A, B with $c_\lambda(A,d) = c_\lambda(B,d)$ for all $\lambda \in \Lambda^+$ and $d \geq 0$, yet $\mathrm{sr}_A(\lambda_0,\mu_0;\nu_0) \neq \mathrm{sr}_B(\lambda_0,\mu_0;\nu_0)$ for some triple (λ_0,μ_0,ν_0).*

Proof. Let $G = \mathrm{GL}_2$ and $M_1 = L(\omega_1)^{\oplus 2}$ (two copies of the standard 2-dimensional representation). Write $L(a,b)$ for $a \geq b$ in GL_2-notation. Then

$$\mathrm{Sym}^2(M_1) = L(2,0)^{\oplus 3} \oplus L(1,1), \qquad \wedge^2(M_1) = L(1,1) \oplus L(0,0).$$

Fix a target module M_2 containing $L(1,1)$ with multiplicity 1 and define a one-parameter family of GL_2-equivariant symmetric bilinear maps $m_t\colon \mathrm{Sym}^2(M_1) \to M_2$ by: m_t acts as the identity on the $L(2,0)$-summands and as multiplication by $t \in \mathbb{C}$ on the $L(1,1)$-summand. By Schur's lemma the parameter t is free.

Define two connected graded GL_2-algebras A and B with $(A_1)_\mu = (B_1)_\mu = (M_1)_\mu$ for all μ, multiplication maps in degree $(1,1)$ given by m_1 (for A) and m_0 (for B), and with all products of total degree ≥ 3 set to zero.

Equal multiplicities. By construction $(A_d)_\mu = (B_d)_\mu$ as vector spaces for all (μ,d): the isotypic dimensions agree in all degrees.

Associativity of B. The map m_0 sets the $L(1,1)$-component of $m_{1,1}$ to zero. In B, all products of total degree ≥ 3 vanish by definition. Every associativity relation $(f \cdot_B g) \cdot_B h = f \cdot_B (g \cdot_B h)$ for $f,g,h \in B_1$ requires checking an equality in $B_3 = 0$; both sides are zero, so associativity holds.

Distinct SRMT values. $\mathrm{sr}_A(L(\omega_1), L(\omega_1); L(1,1)) = 1$ (since m_1 is non-zero on the $L(1,1)$-component), while $\mathrm{sr}_B(L(\omega_1), L(\omega_1); L(1,1)) = 0$ (since m_0 is zero there). □

Remark 3.2 (Scope of Proposition 3.1; orbit-level question)**.** *Proposition 3.1 is a statement about* abstract *connected graded G-algebras. Its role in the GCT context is precise: it shows that the map $A \mapsto \{c_\lambda(A,d)\}_{\lambda,d}$ is not injective on this class, which implies that SRMT-based invariants carry information beyond the BIP obstruction.*

The algebras A and B constructed in the proof are not *coordinate rings of orbit closures; they are truncated symmetric algebras with an artificially zeroed component. The question of whether two G-*orbit closure *coordinate rings can have identical isotypic multiplicities but distinct SRMT values is logically independent and is labeled Open Problem (S4) (Section 9).*

Crucially, the GCT separation argument of Section 9 does not *rely on an orbit-level version of Proposition 3.1. The separation uses:*

(a) Directly computed SRMT values for $\mathbb{C}[O_{\det_m}]$ (Theorems 5.4 and the computation of $\mathrm{sr}^{d_0,d_0}_{O_{\det_3}}(\mu_{\mathrm{perm}},\mu_{\mathrm{perm}};2\mu_{\mathrm{perm}})$ via (S3));

(b) Directly computed SRMT values for $\mathbb{C}[O_{\mathrm{perm}_n}]$ (via (S1) and (S2));

(c) SRMT Monotonicity (Theorem 7.1).

In this argument, the multiplicities of $\mathbb{C}[O_{\mathrm{perm}_n}]$ and $\mathbb{C}[O_{\det_m}]$ at weight μ_{perm} differ (by (S1) and (S3): $c_{\mu_{\mathrm{perm}}}(\mathbb{C}[O_{\mathrm{perm}_n}], d_0) \geq 2$ while $c_{\mu_{\mathrm{perm}}}(\mathbb{C}[O_{\det_m}], d_0) \leq 1$), so the BIP coincidence regime is not invoked. The possible geometric rigidity of orbit closures—whether their geometry could force multiplicities to determine SRMT—is therefore immaterial to the separation argument.

Remark 3.3 (Scenario analysis: what if Open Problem (S1) fails?). *Suppose that* $c_{\mu_{\mathrm{perm}}}(\mathbb{C}[O_{\mathrm{perm}_n}], d) \leq 1$ *for* all $d \geq 0$. *Then:*

(i) The upper bound on the SRMT at $(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}})$ *for any degree pair* (d_0, d_0) *is* $\min(N^{2\mu_{\mathrm{perm}}}_{\mu_{\mathrm{perm}},\mu_{\mathrm{perm}}} \cdot c^2_{\mu_{\mathrm{perm}}}, c_{2\mu_{\mathrm{perm}}}) \leq \min(1 \cdot 1, c_{2\mu_{\mathrm{perm}}}) \leq 1$. *No SRMT value exceeding* 1 *is achievable at this weight triple.*

(ii) The weight triple $(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}})$ *cannot provide a separation of orbit closures via the SRMT. The inequality chain* (3) *collapses.*

(iii) Two responses are available: (a) seek a different weight triple $(\lambda, \mu; \nu)$ *at which* $\mathbb{C}[O_{\mathrm{perm}_n}]$ *and* $\mathbb{C}[O_{\det_m}]$ *exhibit distinct SRMT values; (b) establish* $O_{\mathrm{perm}_n} \not\subseteq O_{\det_m}$ *by other means, e.g., via distinct Hilbert functions, distinct syzygies, or Newton–Okounkov volume inequalities.*

If (S1) fails, the SRMT approach at weight μ_{perm} *is* not *immediately fatal to the general programme: it would require identifying new separation witnesses, but the SRMT framework itself (including Monotonicity and the module–algebra gap) remains valid. In particular, (S1) failing would mean* $\mathbb{C}[O_{\mathrm{perm}_n}]$ *is multiplicity-free at weight* μ_{perm}*—an interesting and unexpected structural fact in its own right.*

Remark 3.4. *In the language of the BIP theorem: the map* $A \mapsto \{c_\lambda(A, d)\}$ *has a non-trivial kernel on the class of* all *graded* G*-algebras, by Proposition 3.1. The SRMT is supported on that kernel. The BIP impossibility result therefore does not obstruct SRMT-based separation arguments for abstract graded* G*-algebras. Whether this kernel intersects the class of orbit-closure coordinate rings non-trivially is Open Problem (S4).*

4 Multiplicity-Free Algebras, Spherical Varieties, and $\{0, 1\}$-SRMT

Definition 4.1 (Multiplicity-free). *A graded* G*-algebra* $A = \bigoplus_{d \geq 0} A_d$ *is* multiplicity-free *if* $c_\lambda(A, d) \leq 1$ *for all* $\lambda \in \Lambda^+$ *and* $d \geq 0$.

Multiplicity-freeness imposes the strongest possible constraint on the SRMT: since $\mathrm{sr}_A(\lambda, \mu; \nu) \leq \min(N^\nu_{\lambda\mu} c_\lambda c_\mu, c_\nu)$ and all multiplicities are at most 1, every SRMT value lies in $\{0, 1\}$.

Theorem 4.2 (Spherical implies multiplicity-free implies $\{0, 1\}$-SRMT). *Let* $O \subset W$ *be a* G*-orbit closure.*

(a) If O *is* G*-spherical (a Borel subgroup* $B \subset G$ *has an open dense orbit in* O*), then* $\mathbb{C}[O]$ *is multiplicity-free as a* G*-module* (Brion [3]).

(b) If $\mathbb{C}[O]$ *is multiplicity-free, then* $\mathrm{sr}_O(\lambda, \mu; \nu) \in \{0, 1\}$ *for all* (λ, μ, ν).

(c) In the multiplicity-free case, $\mathrm{sr}_O(\lambda, \mu; \nu) = 1$ *if and only if* $\nu \in \mathcal{X}(O)$ *and the product map* m_ν *is nonzero.*

Proof. (a) Brion's theorem [3]. (b) Immediately from $\mathrm{sr}_O(\lambda, \mu; \nu) \leq \min(N^\nu_{\lambda\mu} c_\lambda c_\mu, c_\nu) \leq \min(N^\nu_{\lambda\mu}, 1) \in \{0, 1\}$, since all multiplicities are at most 1. (c) Under multiplicity-freeness, the λ- and μ-isotypic components each have unique generators e_λ, e_μ. The SRMT equals 1 if and only if $m_\nu(e_\lambda \otimes e_\mu) \neq 0$ in $A_{d_1+d_2,\nu}$, which is equivalent to $\nu \in \mathcal{X}(O)$ and the map being nonzero. □

Remark 4.3 (Why Brion's theorem does not apply to the H-action on $O_{\det_m}$)**.** *Theorem 4.2(a) applies to a variety that is spherical for the* same *group whose module structure one studies. In our setting we study $\mathbb{C}[O_{\det_m}]$ as an H-module. Brion's theorem would require $O_{\det_m}$ to be spherical for H, i.e., for a Borel subgroup $B_H \subset H$ to have an open dense orbit in $O_{\det_m}$.*

This fails for all $m \geq 2$. Indeed, $\dim O_{\det_m} = \dim G - \dim G_{\det_m}$. The identity component of $G_{\det_m}$ is $(\mathrm{GL}_m \times \mathrm{GL}_m)/\mu_m$, of dimension $2m^2 - 1$ (Remark 2.2), so

$$\dim O_{\det_m} = m^4 - (2m^2 - 1) = (m^2 - 1)^2.$$

A Borel subgroup $B_H \subset H = \mathrm{GL}_m \times \mathrm{GL}_m$ has dimension $\dim B_H = 2 \cdot \frac{m(m+1)}{2} = m(m+1)$.

For $m = 3$: $\dim O_{\det_3} = (9-1)^2 = 64$ and $\dim B_H = 3 \cdot 4 = 12$. Since $12 \ll 64$, no Borel subgroup of H can have an open orbit in $O_{\det_3}$.

For general $m \geq 2$: $(m^2-1)^2 \geq m^4 - 2m^2 + 1$ while $m(m+1) = m^2 + m$; one has $(m^2-1)^2 > m(m+1)$ for all $m \geq 2$ since $m^4 - 4m^2 - m + 1 > 0$ for $m \geq 3$, and for $m = 2$: $(4-1)^2 = 9 > 6 = 2 \cdot 3$. In all cases $\dim O_{\det_m} > \dim B_H$.

Therefore, the hypothesis of Brion's theorem is not satisfied for the H-action on $O_{\det_m}$, and Theorem 4.2(a) cannot be applied to deduce multiplicity-freeness of $\mathbb{C}[O_{\det_m}]$ as an H-module.

Theorem 4.4 (Ray-Multiplicity-1 for the determinant)**.** *For every integer $k \geq 0$,*

$$c_{k\lambda_{\det}}\big(\mathbb{C}[O_{\det_m}], k\big) = 1.$$

Equivalently, the $k\lambda_{\det}$-isotypic subspace of $\mathbb{C}[O_{\det_m}]_k$ is one-dimensional, spanned by $(\det_m)^k$ (in the sense of Remark 2.10).

Proof. **Step 1 (Upper bound).** The weight $\lambda_{\det} = (\omega_m; \omega_m)$ is the determinant character of H, so $L^H(\lambda_{\det}) = \bigwedge^m \mathbb{C}^m \otimes \bigwedge^m \mathbb{C}^m$ has dimension 1. For every $k \geq 0$, the module $L^H(k\lambda_{\det}) = (L^H(\lambda_{\det}))^{\otimes k}$ has dimension 1. Hence any H-submodule of isotypic type $k\lambda_{\det}$ in any H-module has dimension at most 1, giving $c_{k\lambda_{\det}}(\mathbb{C}[O_{\det_m}], k) \leq 1$.

Step 2 (Non-vanishing of $(\det_m)^k$). By Remark 2.10, $(\det_m)^k = \widehat{\det}_m^{\,k} \in A_k$ is the k-th power of the degree-1 element of H-weight $\lambda_{\det}$, hence of H-weight $k\lambda_{\det}$ in A_k. To confirm non-vanishing, evaluate at the identity matrix Id_m: since $\det_m(\mathrm{Id}_m) = 1 \neq 0$, the point Id_m lies in $O_{\det_m}$, and

$$\mathrm{ev}_{\mathrm{Id}}\big((\det_m)^k\big) = \big(\det_m(\mathrm{Id}_m)\big)^k = 1 \neq 0.$$

Hence $(\det_m)^k \neq 0$ in $\mathbb{C}[O_{\det_m}]_k$, giving $c_{k\lambda_{\det}}(A, k) \geq 1$.

Conclusion. $c_{k\lambda_{\det}}(\mathbb{C}[O_{\det_m}], k) = 1$ for all $k \geq 0$. □

Remark 4.5. *Theorem 4.4 is strictly weaker than full multiplicity-freeness: it asserts $c_{k\lambda_{\det}} = 1$ only for the specific weights along the ray $\{k\lambda_{\det}\}_{k\geq 0}$, not for arbitrary H-dominant weights. This is, however, precisely the information required for Bilinear Rank-1 Rigidity (Theorem 5.4) and all subsequent unconditional results.*

Conjecture 4.6 (Full multiplicity-freeness of the determinant orbit closure)**.** *The coordinate ring $\mathbb{C}[O_{\det_m}]$ is multiplicity-free as an $H = \mathrm{GL}_m \times \mathrm{GL}_m$-module; equivalently, $c_\mu(\mathbb{C}[O_{\det_m}], d) \leq 1$ for all $\mu \in \Lambda^+(H)$ and $d \geq 0$.*

Remark 4.7 (On Conjecture 4.6)**.** *A proof of this conjecture would require either: (a) identifying a group G' with $H \subset G' \subset G$ for which $O_{\det_m}$ is G'-spherical and $\mathbb{C}[O_{\det_m}]$ is multiplicity-free as a G'-module (and hence as an H-module by restriction), or (b) a direct structural analysis of $\mathbb{C}[O_{\det_m}]$ using the geometry of determinantal varieties and the equivariant resolution theory of Weyman [11].*

Even the special case $m = 3$ (where $\dim O_{\det_3} = 64$) would represent meaningful progress, accessible via Macaulay2. For the separation programme, only Open Problem (S3)—which requires $c_{2\mu_{\mathrm{perm}}}(\mathbb{C}[O_{\det_3}]) = 1$—is used, and this specific multiplicity can be verified computationally without proving the full conjecture.

5 Multiplicativity and SRMT Rigidity of the Determinant

Definition 5.1 (Multiplicative polynomial)**.** *A polynomial $f \in W$ is (H, χ)-*multiplicative *for a subgroup $H \leq G$ and character $\chi\colon H \to \mathbb{C}^*$ if $f(h \cdot x) = \chi(h) f(x)$ for all $h \in H$ and $x \in \mathbb{C}^{m^2}$.*

The fundamental example is $\det_m(A^T X B) = \det(A)\det(B)\det_m(X)$, i.e., $\det_m$ is (H, χ)-multiplicative with $\chi(A, B) = \det(A)\det(B)$.

Lemma 5.2 (Multiplicativity Rigidity)**.** *Let $f \in W$ be (H, χ)-multiplicative with H-weight $\lambda_{\det} = \chi$ a one-dimensional character of H. Assume $c_{\lambda_{\det}}(\mathbb{C}[O_f], 1) = 1$. Then*

$$\mathrm{sr}^H_{O_f}(k\lambda_{\det}, k\lambda_{\det}; 2k\lambda_{\det}) = 1 \quad \textit{for all } k \geq 1.$$

Proof. Case $k = 1$. Since $c_{\lambda_{\det}}(\mathbb{C}[O_f], 1) = 1$, the $\lambda_{\det}$-isotypic component of $\mathbb{C}[O_f]_1$ is $\mathrm{span}(\hat{f})$. The product $\hat{f} \cdot \hat{f} \in A_2$ is an H-eigenvector of weight $2\lambda_{\det}$. Since $\lambda_{\det}$ is a one-dimensional character, $L^H(2\lambda_{\det})$ is one-dimensional, so $c_{2\lambda_{\det}}(A, 2) \leq 1$. The element $\hat{f}^2$ is nonzero in A_2 (by the same evaluation argument as in Theorem 4.4, Step 2), so $c_{2\lambda_{\det}}(A, 2) = 1$. Hence $m_{2\lambda_{\det}}$ is a linear map between one-dimensional spaces sending $\hat{f} \otimes \hat{f} \mapsto \hat{f}^2 \neq 0$, an isomorphism of rank 1.

Case $k \geq 2$. The same argument applies with $\hat{f}^k$ and $\hat{f}^{2k}$: both are nonzero H-eigenvectors spanning their respective one-dimensional isotypic components (by Theorem 4.4), and the product map has rank 1. □

Theorem 5.3 (Rank-1 Rigidity along the diagonal ray)**.** *$c_{\lambda_{\det}}(O_{\det_m}, 1) = 1$ and $\mathrm{sr}^H_{O_{\det_m}}(k\lambda_{\det}, k\lambda_{\det}; 2k\lambda_{\det}) = 1$ for all $k \geq 1$.*

Proof. By Theorem 4.4, $c_{\lambda_{\det}}(\mathbb{C}[O_{\det_m}], 1) = 1$. Since $\det_m$ is $(H, \det(A)\det(B))$-multiplicative and $\lambda_{\det}$ is a one-dimensional character, Lemma 5.2 applies. □

Theorem 5.4 (Bilinear Rank-1 Rigidity)**.** *For all integers $a, b \geq 0$ with $a + b \geq 1$,*

$$\mathrm{sr}^H_{O_{\det_m}}(a\lambda_{\det}, b\lambda_{\det}; (a+b)\lambda_{\det}) = 1.$$

Equivalently, the isotypic multiplication map

$$m_{(a+b)\lambda_{\det}}\colon \big(\mathbb{C}[O_{\det_m}]_a\big)_{a\lambda_{\det}} \otimes \big(\mathbb{C}[O_{\det_m}]_b\big)_{b\lambda_{\det}} \longrightarrow \big(\mathbb{C}[O_{\det_m}]_{a+b}\big)_{(a+b)\lambda_{\det}}$$

is an isomorphism of one-dimensional $\mathbb{C}$-vector spaces for every $a, b \geq 0$ with $a + b \geq 1$.

Proof. **Step 1 (Multiplicities along the ray).** By Theorem 4.4: $c_{a\lambda_{\det}}(\mathbb{C}[O_{\det_m}], a) = c_{b\lambda_{\det}}(\mathbb{C}[O_{\det_m}], b) = c_{(a+b)\lambda_{\det}}(\mathbb{C}[O_{\det_m}], a+b) = 1$. For each $k \geq 0$, the $k\lambda_{\det}$-isotypic component of A_k is $\mathbb{C} \cdot (\det_m)^k$.

Step 2 (LR coefficient). Since $\lambda_{\det} = (\omega_m; \omega_m)$ is a one-dimensional character, $L^H(a\lambda_{\det}) \otimes L^H(b\lambda_{\det}) = L^H((a+b)\lambda_{\det})$, so $N^{(a+b)\lambda_{\det}}_{a\lambda_{\det}, b\lambda_{\det}} = 1$ (Remark 2.12).

Step 3 (The map is an isomorphism). The maximum possible SRMT value at $(a\lambda_{\det}, b\lambda_{\det}; (a+b)\lambda_{\det})$ is $\min(1{\cdot}1{\cdot}1, 1) = 1$. The map $m_{(a+b)\lambda_{\det}}$ sends $(\det_m)^a \otimes (\det_m)^b \mapsto (\det_m)^{a+b}$, which is nonzero in A_{a+b} (Theorem 4.4, Step 2). Hence the rank is 1. □

Remark 5.5 (Source of Rank-1 Rigidity)**.** *The true engine of Theorem 5.4 is the one-dimensionality of $L^H(k\lambda_{\det})$ for every $k \geq 0$. Since $\lambda_{\det} = (\omega_m; \omega_m)$ is the determinant character, $L^H(a\lambda_{\det}) \otimes L^H(b\lambda_{\det}) = L^H((a+b)\lambda_{\det})$ is one-dimensional. This forces $m_{(a+b)\lambda_{\det}}$ to be a linear map between one-dimensional spaces, so its rank is at most 1; non-vanishing of $(\det_m)^{a+b}$ promotes this to exactly 1. Neither Brion's theorem nor any sphericity hypothesis is invoked.*

Contrast with the permanent: $L^H(k\mu_{\mathrm{perm}}) = \mathrm{Sym}^{kn}(\mathbb{C}^n) \otimes \mathrm{Sym}^{kn}(\mathbb{C}^n)$ has dimension $\binom{kn+n-1}{n}^2 \gg 1$, so no analogous dimensional forcing exists.

Corollary 5.6 (Polynomial sub-algebra along the $\lambda_{\det}$-ray)**.** *The H-isotypic sub-algebra $R_{\lambda_{\det}}(O_{\det_m}) := \bigoplus_{k \geq 0} (\mathbb{C}[O_{\det_m}]_k)_{k\lambda_{\det}}$ is isomorphic, as a graded $\mathbb{C}$-algebra, to the polynomial ring $\mathbb{C}[t]$ with $\deg t = 1$.*

Proof. By Theorem 4.4, $(\mathbb{C}[O_{\det_m}]_k)_{k\lambda_{\det}} = \mathbb{C} \cdot (\det_m)^k$ for every $k \geq 0$. Theorem 5.4 shows the product map is an isomorphism of one-dimensional graded pieces. Hence $R_{\lambda_{\det}}(O_{\det_m})$ is freely generated by $\det_m$ with no polynomial relations: $R_{\lambda_{\det}}(O_{\det_m}) \cong \mathbb{C}[t]$ via $t \mapsto \det_m$. □

Theorem 5.7 (Symmetries of the permanent and the dimension obstruction)**.** *Let $n \geq 2$ and $T \cong (\mathbb{C}^*)^{2n-1}$ be the diagonal torus of row/column rescalings modulo the global scalar (the identity component of $\mathrm{Stab}_{\mathrm{GL}_{n^2}}(\mathrm{perm}_n)$).*

(i) (Torus acts by character.) *Each monomial $\prod_i x_{i,\sigma(i)}$ in perm_n is scaled by $(\prod_i t_i)(\prod_j s_j)$ uniformly over all $\sigma \in S_n$, so perm_n is a T-eigenvector with character $\chi_T(t) = \prod t_i \cdot \prod s_j$.*

(ii) (The weight space is high-dimensional.) $\dim L^H(\mu_{\mathrm{perm}}) = \dim(\mathrm{Sym}^n(\mathbb{C}^n) \otimes \mathrm{Sym}^n(\mathbb{C}^n)) = \binom{2n-1}{n}^2 \gg 1$ *for $n \geq 2$. In particular: $n = 2 \Rightarrow \binom{3}{2}^2 = 9$; $n = 3 \Rightarrow \binom{5}{3}^2 = 100$.*

(iii) (Dimension obstruction to rank-1 rigidity.) *Because $\dim L^H(\mu_{\mathrm{perm}}) > 1$, the μ_{perm}-isotypic component of any H-module can have multiplicity ≥ 1 without any character identity forcing that multiplicity to exactly 1. No analogue of Bilinear Rank-1 Rigidity can be established for $\mathbb{C}[O_{\mathrm{perm}_n}]$ from first principles.*

Proof. (i) Each monomial $\prod_i x_{i,\sigma(i)}$ scales by $\prod_i t_i \cdot \prod_j s_j$, independently of σ. (ii) The Cauchy decomposition places perm_n in $\mathrm{Sym}^n(\mathbb{C}^{n*}) \otimes \mathrm{Sym}^n(\mathbb{C}^{n*})$. (iii) The proof of Theorem 5.4 relied critically on $\dim L^H(\lambda_{\det}) = 1$. Since $\dim L^H(\mu_{\mathrm{perm}}) > 1$, multiplicities $c_{k\mu_{\mathrm{perm}}}$ are not bounded by 1, and every SRMT value at weight triples $(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}})$ is unconstrained. □

Example 5.8 (The permanent and the full group H)**.** *The torus T acts on perm_n by a character, but the full group H does not. For $n = 2$: take $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $B = I_2$. Then*

$\mathrm{perm}_2(AX) = \mathrm{perm}_2(X) + 2x_{21}x_{22} \neq \chi(A,B)\mathrm{perm}_2(X)$ *for generic* X*, confirming that the character action is restricted to the torus* T*.*

Remark 5.9 (Failure of Rank-1 Rigidity for perm_n)**.** *Bilinear Rank-1 Rigidity rests on* $\dim L^H(\lambda_{\mathrm{det}}) = 1$*. For* μ_{perm}*, with* $m \geq 2$*,* $\dim L^H(\mu_{\mathrm{perm}}) = \binom{m+m-1}{m}^2 > 1$*. Also,* $N^{2\mu_{\mathrm{perm}}}_{\mu_{\mathrm{perm}},\mu_{\mathrm{perm}}} = 1$ *(Pieri rule; Remark 2.12). Consequently:*

(i) The SRMT bound at $(a\mu_{\mathrm{perm}}, b\mu_{\mathrm{perm}}; (a+b)\mu_{\mathrm{perm}})$ *is* $\min(c_{a\mu_{\mathrm{perm}}} \cdot c_{b\mu_{\mathrm{perm}}}, c_{(a+b)\mu_{\mathrm{perm}}})$ *(using* $N = 1$*). If* $c_{\mu_{\mathrm{perm}}} \geq 2$ *(Open Problem (S1)), the bound is* ≥ 2 *and SRMT values* ≥ 2 *are not a priori excluded.*

(ii) The multiplicity $c_{k\mu_{\mathrm{perm}}}(\mathbb{C}[O_{\mathrm{perm}_n}], k)$ *is not forced to* ≤ 1 *by any character identity.*

(iii) The analogue of Corollary 5.6 fails for $\mathbb{C}[O_{\mathrm{perm}_n}]$ *in general.*

The flexibility of the SRMT for the permanent is conditional on (S1).

6 SRMT Defects and the Structure of Syzygies

Definition 6.1 (Isotypic syzygy and cosyzygy)**.** *For a graded* G*-algebra* A*, fixed degrees* $d_1, d_2 \geq 1$*, and weights* $\lambda, \mu, \nu \in \Lambda^+$*:*

$$\mathrm{Syz}^{d_1,d_2}(A)_{\lambda,\mu;\nu} := \ker\big(m_\nu \colon (A_{d_1})_\lambda \otimes (A_{d_2})_\mu \to A_{d_1+d_2,\nu}\big),$$

$$\mathrm{CoSyz}^{d_1,d_2}(A)_{\lambda,\mu;\nu} := \mathrm{coker}\big(m_\nu \colon (A_{d_1})_\lambda \otimes (A_{d_2})_\mu \to A_{d_1+d_2,\nu}\big).$$

Theorem 6.2 (SRMT–Syzygy exact sequence)**.** *For any graded* G*-algebra* A *and* $\lambda, \mu, \nu \in \Lambda^+$*:*

$$0 \to \mathrm{Syz}^{d_1,d_2}(A)_{\lambda,\mu;\nu} \to (A_{d_1})_\lambda \otimes (A_{d_2})_\mu \xrightarrow{m_\nu} A_{d_1+d_2,\nu} \to \mathrm{CoSyz}^{d_1,d_2}(A)_{\lambda,\mu;\nu} \to 0,$$

with

$$\mathrm{sr}^{d_1,d_2}_A(\lambda,\mu;\nu) = N^\nu_{\lambda\mu}\, c_\lambda(A,d_1)\, c_\mu(A,d_2) - \dim \mathrm{Syz}^{d_1,d_2}(A)_{\lambda,\mu;\nu} = c_\nu(A,d_1+d_2) - \dim \mathrm{CoSyz}^{d_1,d_2}(A)_{\lambda,\mu;\nu}.$$

Proof. Standard rank-nullity for m_ν, using that $(A_{d_1})_\lambda \otimes (A_{d_2})_\mu$ has dimension $N^\nu_{\lambda\mu} c_\lambda c_\mu$ and $A_{d_1+d_2,\nu}$ has dimension c_ν. □

Theorem 6.3 (Rigidity criterion for SRMT maximality)**.** $\mathrm{sr}^{d_1,d_2}_A(\lambda,\mu;\nu) = \min(N^\nu_{\lambda\mu} c_\lambda c_\mu, c_\nu)$ *if and only if either:*

(a) $\mathrm{Syz}^{d_1,d_2}(A)_{\lambda,\mu;\nu} = 0$ *(injectivity regime), or*

(b) $\mathrm{CoSyz}^{d_1,d_2}(A)_{\lambda,\mu;\nu} = 0$ *(surjectivity regime).*

Proof. Immediate from Theorem 6.2 and rank-nullity. □

Corollary 6.4 (SRMT under global degree-1 generation)**.** *Let* $A = \bigoplus_{d\geq 0} A_d$ *be a connected graded* G*-algebra generated in degree* 1*. Then the global multiplication map* $m_{1,1} \colon A_1 \otimes A_1 \twoheadrightarrow A_2$ *is surjective. At the level of* ν*-isotypic components, for each* $\nu \in \Lambda^+$*:*

$$A_{2,\nu} = \sum_{(\lambda,\mu):N^\nu_{\lambda\mu}>0} \mathrm{Im}\big(m_\nu \colon (A_1)_\lambda \otimes (A_1)_\mu \to A_{2,\nu}\big). \tag{2}$$

(i) *By subadditivity:* $c_\nu(A,2) \le \sum_{(\lambda,\mu):N^\nu_{\lambda\mu}>0} \mathrm{sr}^{1,1}_A(\lambda,\mu;\nu)$*, with equality iff the images are linearly independent.*

(ii) *For a fixed pair* (λ_0,μ_0)*, the cosyzygy* $\mathrm{CoSyz}^{1,1}(A)_{\lambda_0,\mu_0;\nu}$ *need not vanish; degree-*1 *generation does not imply individual surjectivity at* (λ_0,μ_0)*.*

(iii) *Independently of any surjectivity assumption:* $\dim \mathrm{Syz}^{1,1}(A)_{\lambda_0,\mu_0;\nu} = N^\nu_{\lambda_0\mu_0}\, c_{\lambda_0}(A,1)\, c_{\mu_0}(A,1) - \mathrm{sr}^{1,1}_A(\lambda_0,\mu_0;\nu)$*.*

Remark 6.5 (Correction to an earlier version)**.** *A previous draft stated that generation in degree* 1 *implies* $\mathrm{CoSyz}^{1,1}(A)_{\lambda,\mu;\nu} = 0$ *for each individual pair* (λ,μ)*. This was an error. Global surjectivity of* $m_{1,1}$ *implies the* joint *surjectivity* (2)*, not individual surjectivity at any single pair. The individual SRMT defect* $\delta^{1,1}_A(\lambda,\mu;\nu)$ *can be positive even when* A *is generated in degree* 1*.*

7 SRMT Monotonicity

Theorem 7.1 (SRMT Monotonicity at fixed degree pairs)**.** *Let* $O_1 \subseteq O_2$ *be* G*-orbit closures in* W*, and let* $d_1, d_2 \ge 0$ *be fixed integers. Then*

$$\mathrm{sr}^{d_1,d_2}_{O_1}(\lambda,\mu;\nu) \le \mathrm{sr}^{d_1,d_2}_{O_2}(\lambda,\mu;\nu) \quad \textit{for all } \lambda,\mu,\nu \in \Lambda^+.$$

Proof. See Appendix C. □

Remark 7.2. *Monotonicity holds for each fixed degree pair* (d_1,d_2)*. Comparison at respective minimal degrees involves different degree pairs and falls outside this theorem; see Remark 2.4.*

Corollary 7.3. *If* $\mathrm{sr}^{d_1,d_2}_{O_1}(\lambda_0,\mu_0;\nu_0) > \mathrm{sr}^{d_1,d_2}_{O_2}(\lambda_0,\mu_0;\nu_0)$ *for a single triple* $(\lambda_0,\mu_0;\nu_0)$ *and fixed* (d_1,d_2)*, then* $O_1 \not\subseteq O_2$*.*

8 The Permanent's SRMT

Proposition 8.1 (Non-sphericity of O_{perm_n} for $n \ge 2$)**.** *For all* $n \ge 2$*, the orbit* $G \cdot \mathrm{perm}_n$ *is not spherical for* $G = \mathrm{GL}_{n^2}$*. Consequently, Brion's criterion for multiplicity-freeness of* $\mathbb{C}[O_{\mathrm{perm}_n}]$ *as a* G*-module is not satisfied.*

Proof. The stabiliser of perm_n in $G = \mathrm{GL}_{n^2}$ contains the group $(S_n \times S_n) \ltimes (\mathbb{C}^*)^{2n-1}$, where $S_n \times S_n$ acts by simultaneous permutation of rows and columns, and $(\mathbb{C}^*)^{2n-1}$ is the torus of row/column rescalings modulo the global scalar (Theorem 5.7). The identity component of G_{perm_n} is therefore $(\mathbb{C}^*)^{2n-1}$, of dimension $2n-1$, giving

$$\dim(G \cdot \mathrm{perm}_n) = n^4 - (2n-1).$$

A Borel subgroup $B_{n^2} \subset \mathrm{GL}_{n^2}$ has dimension $\dim B_{n^2} = \frac{n^2(n^2+1)}{2}$. We compare these for small values of n:

n	$\dim(G \cdot \mathrm{perm}_n) = n^4 - (2n-1)$	$\dim B_{n^2} = \frac{n^2(n^2+1)}{2}$	spherical?
2	$16 - 3 = 13$	$\frac{4\cdot 5}{2} = 10$	No ($13 > 10$)
3	$81 - 5 = 76$	$\frac{9\cdot 10}{2} = 45$	No ($76 > 45$)
4	$256 - 7 = 249$	$\frac{16\cdot 17}{2} = 136$	No ($249 > 136$)

For general $n \geq 2$, we verify $n^4 - (2n-1) > \frac{n^2(n^2+1)}{2}$, i.e., $2n^4 - 4n + 2 > n^4 + n^2$, i.e., $n^4 - n^2 - 4n + 2 > 0$. For $n \geq 2$: $n^4 - n^2 - 4n + 2 \geq 16 - 4 - 8 + 2 = 6 > 0$. The inequality is strict for all $n \geq 2$.

Since $\dim(G \cdot \mathrm{perm}_n) > \dim B_{n^2}$ for all $n \geq 2$, no Borel subgroup of GL_{n^2} has an open orbit in $G \cdot \mathrm{perm}_n$, so the orbit is not spherical. □

Remark 8.2 (Non-sphericity versus non-multiplicity-freeness)**.** *Proposition 8.1 establishes that the sufficient condition for multiplicity-freeness given by Brion's theorem is not fulfilled. The converse of Brion's theorem is false in general (non-spherical does not imply not multiplicity-free). Whether $\mathbb{C}[O_{\mathrm{perm}_n}]$ actually has some $c_\lambda \geq 2$ is a separate question; this is Open Problem (S1).*

Remark 8.3 (SRMT flexibility for the permanent, conditional on (S1))**.** *Suppose $c_{\mu_{\mathrm{perm}}}(\mathbb{C}[O_{\mathrm{perm}_n}], d_0) \geq 2$ for some degree d_0 (Open Problem (S1)). Then the maximum possible SRMT at $(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}})$ at degree pair (d_0, d_0) is $\min(c^2_{\mu_{\mathrm{perm}}}, c_{2\mu_{\mathrm{perm}}}) \geq 2$ (using $N^{2\mu_{\mathrm{perm}}}_{\mu_{\mathrm{perm}},\mu_{\mathrm{perm}}} = 1$). Since no character identity constrains the multiplication maps (Theorem 5.7), there is no algebraic mechanism forcing the SRMT below its theoretical upper bound.*

Theorem 8.4 (Structural asymmetry between determinant and permanent)**.** *Let $m \geq n \geq 2$.*

(i) (Unconditional.) *For every $a, b \geq 0$ with $a+b \geq 1$: $\mathrm{sr}^H_{O_{\det_m}}(a\lambda_{\det}, b\lambda_{\det}; (a+b)\lambda_{\det}) = 1$. No analogue of Bilinear Rank-1 Rigidity can be derived for $\mathbb{C}[O_{\mathrm{perm}_n}]$ from first principles, due to $\dim L^H(\mu_{\mathrm{perm}}) \gg 1$.*

(ii) (Unconditional.) *The orbit $G \cdot \mathrm{perm}_n$ is not spherical for $G = \mathrm{GL}_{n^2}$; Brion's multiplicity-free criterion fails for $\mathbb{C}[O_{\mathrm{perm}_n}]$.*

(iii) (Conditional on (S1)+(S2)). *If $c_{\mu_{\mathrm{perm}}}(\mathbb{C}[O_{\mathrm{perm}_n}], d_0) \geq 2$ and the multiplication matrix has rank ≥ 2, then $\mathrm{sr}^{d_0,d_0}_{O_{\mathrm{perm}_n}}(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}}) \geq 2$.*

(iv) (Unconditional.) *SRMT values are monotone under orbit-closure inclusion at any fixed degree pair.*

9 The $n = 3$ Specialisation and the Computational Programme

For $n = m = 3$ all ingredients are finite-dimensional: $G = \mathrm{GL}_9$, $W = \mathrm{Sym}_3(\mathbb{C}^{9*})$ ($\dim W = \binom{9+2}{3} = 165$). The Cauchy decomposition gives three H-components:

Component λ	H-weight	Contains	dim
(1^3): $\bigwedge^3 U^* \otimes \bigwedge^3 V^*$	$\lambda_{\det} = (\omega_3; \omega_3)$	$\det_3$	1
$(2,1)$: $S_{(2,1)}(U^*) \otimes S_{(2,1)}(V^*)$	$((2,1,0); (2,1,0))$	neither	64
(3): $\mathrm{Sym}^3 U^* \otimes \mathrm{Sym}^3 V^*$	$\mu_{\mathrm{perm}} = (3\omega_1; 3\omega_1)$	perm_3	100

9.1 The structural gap at $n = 3$

The proved unconditional results specialise as follows:

- $\mathrm{sr}^H_{O_{\det_3}}(a\lambda_{\det}, b\lambda_{\det}; (a+b)\lambda_{\det}) = 1$ for all $a, b \geq 0$, $a + b \geq 1$ (Theorem 5.4).
- $G \cdot \mathrm{perm}_3$ is not spherical for $G = \mathrm{GL}_9$ (Proposition 8.1).
- No character identity constrains the SRMT of $\mathbb{C}[O_{\mathrm{perm}_3}]$ (Theorem 5.7).

9.2 Open problems: the computational separation programme

Open Problem (S1). *Non-trivial multiplicity for the permanent.* Compute the H-isotypic decomposition of $\mathbb{C}[O_{\mathrm{perm}_3}]_d$ for $d = 1, 2, 3, \ldots$ Determine the smallest degree $d_0 \geq 2$ at which $c_{\mu_{\mathrm{perm}}}(\mathbb{C}[O_{\mathrm{perm}_3}], d_0) > 0$, and verify whether $c_{\mu_{\mathrm{perm}}}(\mathbb{C}[O_{\mathrm{perm}_3}], d_0) \geq 2$. *Expected: yes; status: open and computable; see Remark 9.2.*

Open Problem (S2). *SRMT value ≥ 2 for the permanent.* Set $\nu^* = 2\mu_{\mathrm{perm}}$. Assuming (S1) holds with $r := c_{\mu_{\mathrm{perm}}}(\mathbb{C}[O_{\mathrm{perm}_3}], d_0) \geq 2$, construct the $r \times r$ multiplication matrix $M_{\mu_{\mathrm{perm}},\mu_{\mathrm{perm}};\nu^*} := (m_{\nu^*}(v_i \otimes v_j))^r_{i,j=1}$, where $\{v_i\}$ is a basis for $(\mathbb{C}[O_{\mathrm{perm}_3}]_{d_0})_{\mu_{\mathrm{perm}}}$, and verify $\operatorname{rank} M \geq 2$. *Expected: yes; status: open, depends on (S1).*

Open Problem (S3). *SRMT value ≤ 1 for the determinant at $(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; \nu^*)$.* Compute $c_{\nu^*}(\mathbb{C}[O_{\det_3}])$ for $\nu^* = 2\mu_{\mathrm{perm}} = (6\omega_1; 6\omega_1)$ directly from $I(O_{\det_3})$. Since $\mu_{\mathrm{perm}} \neq k\lambda_{\det}$ (Remark 2.11), this does not follow automatically from Theorem 5.4. Expected: $c_{\nu^*}(\mathbb{C}[O_{\det_3}]) = 1$, consistent with Conjecture 4.6; to be verified independently. *Status: open.*

Open Problem (S4). *Orbit-level multiplicity–SRMT independence.* Do there exist G-orbit closures $O_1, O_2 \subset W$ with $c_\lambda(\mathbb{C}[O_1], d) = c_\lambda(\mathbb{C}[O_2], d)$ for all (λ, d), yet $\mathrm{sr}^{d_1,d_2}_{O_1}(\lambda, \mu; \nu) \neq \mathrm{sr}^{d_1,d_2}_{O_2}(\lambda, \mu; \nu)$ for some $(\lambda, \mu, \nu, d_1, d_2)$? A positive answer would extend the scope of SRMT-based separation to the BIP coincidence regime (Remark 2.5). A negative answer—multiplicities determine SRMT for orbit closures—would mean geometric rigidity imposes a stronger constraint than exists for abstract graded algebras, and would require a fundamentally different separation strategy. *Status: open; does not affect the separation via (S1)–(S3).*

Logical conclusion. If (S1), (S2), and (S3) all hold, then by SRMT Monotonicity (Corollary 7.3) at degree pair (d_0, d_0):

$$\mathrm{sr}^{d_0,d_0}_{O_{\mathrm{perm}_3}}(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; \nu^*) \geq 2 > 1 \geq \mathrm{sr}^{d_0,d_0}_{O_{\det_3}}(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; \nu^*), \tag{3}$$

which gives $O_{\mathrm{perm}_3} \not\subseteq O_{\det_3}$ and hence $\mathbf{VP} \neq \mathbf{VNP}$.

Remark 9.1 (Logical structure of the separation argument)**.** *The inequality chain* (3) *relies on all three primary open problems:*

- ***(S1)*** *is necessary for the left-hand side ≥ 2: the SRMT bound at $(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; 2\mu_{\mathrm{perm}})$ is $\min(c^2_{\mu_{\mathrm{perm}}}, c_{2\mu_{\mathrm{perm}}})$ (using $N = 1$); for this to exceed 1 one needs $c_{\mu_{\mathrm{perm}}}(\mathbb{C}[O_{\mathrm{perm}_3}], d_0) \geq 2$.*
- ***(S2)*** *verifies that no accidental algebraic cancellation reduces the multiplication matrix rank below its theoretical upper bound.*

- ***(S3)*** *is necessary and does not follow from Theorem 5.4: since* $\mu_{\mathrm{perm}} \neq k\lambda_{\mathrm{det}}$ *(Remark 2.11), the weight triple* $(\mu_{\mathrm{perm}}, \mu_{\mathrm{perm}}; \nu^*)$ *lies off the* λ_{det}*-ray. Bilinear Rank-*1 *Rigidity controls SRMT values only along that ray.*

The three conditions are logically independent; none implies any other.

Remark 9.2 (Computational feasibility of (S1)–(S3))**.** *We assess the feasibility of the three open problems using current algorithms.*

(S1): Isotypic multiplicities. *The most tractable approach does not rely on computing* $I(O_{\mathrm{perm}_3})$ *via parametric elimination. Instead, one proceeds as follows:*

- *The ambient space* $W = \mathrm{Sym}_3(\mathbb{C}^{9*})$ *has dimension* 165*; the group* $G = \mathrm{GL}_9$ *acts on it.*

- *Compute the isotypic decomposition of* $\mathbb{C}[O_{\mathrm{perm}_3}]_d$ *using the Hilbert series and character methods available in Macaulay2's* `SchurRings` *package, without explicitly constructing* $I(O_{\mathrm{perm}_3})$*.*

- *Alternatively, compute the character of* $\mathbb{C}[O_{\mathrm{perm}_3}]_d$ *as a* GL_9 *representation by the equivariant Hilbert series formula, then decompose using the Weyl character formula. This is a finite linear algebra computation in the* 165*-dimensional space* W *and is feasible for small* d *(e.g.,* $d \leq 4$*).*

Gröbner basis approach (for reference). *The parametric elimination strategy—introduce* 81 *variables for* $g \in \mathrm{GL}_9$*, write the action* $g \cdot \mathrm{perm}_3$ *as a polynomial in* 9 *variables, then eliminate the* 81 *group variables—leads to a Gröbner basis computation in* $81+9 = 90$ *variables with the Eliminate order. For* $\dim O_{\mathrm{perm}_3} = 76$*, this is infeasible with current Gröbner basis implementations: such an elimination is known to require doubly-exponential time in the number of elimination variables and the degree. The Macaulay2 scripts in Appendix B are therefore sketched as conceptual programs; their actual execution requires either the character-theoretic approach above or specialised algorithms (e.g., those exploiting the* H*-equivariance to reduce the computation).*

(S2): Rank computation. *Once (S1) is resolved and a basis* $\{v_i\}$ *for the* μ_{perm}*-isotypic component is obtained, computing the rank of the* $r \times r$ *matrix* M *is elementary linear algebra, feasible in Macaulay2 over* $\mathbb{Q}$*.*

(S3): Isotypic decomposition of $\mathbb{C}[O_{\mathrm{det}_3}]$***.*** *The variety* O_{det_3} *has* $\dim = 64$ *and is related to (but not equal to) classical determinantal varieties. Its coordinate ring* $\mathbb{C}[O_{\mathrm{det}_3}]$ *is accessible via the equivariant Hilbert series and* `SchurRings` *computations, as in (S1). The weight* $\nu^* = 2\mu_{\mathrm{perm}} = (6\omega_1; 6\omega_1)$ *requires decomposing* $\mathbb{C}[O_{\mathrm{det}_3}]_d$ *at the appropriate degree, which is a finite computation given the Hilbert series of* O_{det_3} *(known from the Hochster–Eagon resolution [7] for related determinantal varieties).*

10 Perspectives

10.1 Koszul structure of $\mathbb{C}[O_{\mathrm{det}_m}]$

Definition 10.1 (Koszul algebra; Priddy [8])**.** *A connected graded* $\mathbb{C}$*-algebra* $A = \bigoplus_{d \geq 0} A_d$*, generated in degree* 1*, with* $A_0 = \mathbb{C}$*, is* Koszul *if* $\mathrm{Ext}^i_A(\mathbb{C}, \mathbb{C})_j = 0$ *for all* $i \neq j$*.*

Conjecture 10.2 (H-equivariant Koszul property)**.** *Assume the following two prerequisites hold for* $\mathbb{C}[O_{\det_m}]$*:*

(K1) $\mathbb{C}[O_{\det_m}]$ *is generated in degree* 1 *as a graded* $\mathbb{C}$*-algebra.*

(K2) The ideal $I(O_{\det_m})$ *is generated in degree* 2*.*

Then $\mathbb{C}[O_{\det_m}]$ *is* H*-equivariantly Koszul, with the Koszul dual algebra* $A^! = \mathrm{Ext}^*_A(\mathbb{C}, \mathbb{C})$ *carrying a natural* H*-module structure. The ideal of relations in degree* 2 *is generated, as an* H*-equivariant submodule of* $\mathrm{Sym}^2(A_1)$*, by the kernel of the surjection*

$$\mathrm{Sym}^2(V_{\omega_m} \boxtimes V_{\omega_m}) \twoheadrightarrow V_{2\omega_m} \boxtimes V_{2\omega_m},$$

controlled by Plücker-type quadratic relations arising from the Cauchy decomposition.

Remark 10.3 (Status, limitations, and syzygy analysis for hypotheses (K1) and (K2))**.** *Both prerequisites of Conjecture 10.2 are unverified hypotheses.*

(K1): Generation in degree 1**.** *The variety* $O_{\det_m}$ *is an affine cone in* W*, so its coordinate ring is a quotient of* $\mathbb{C}[W]$*, generated in degree* 1*. Generation in degree* 1 *for the quotient is expected by analogy with classical determinantal varieties (for which the Hochster–Eagon theorem [7] implies Cohen–Macaulay structure, compatible with degree-*1 *generation) but requires verification for the orbit closure* $O_{\det_m}$ *inside* $W = \mathrm{Sym}_m(\mathbb{C}^{m^2*})$*, which is a different variety.*

(K2): Degree-*2 *generation of $I(O_{\det_m})$**.** *This is the critical and substantially more speculative hypothesis. We provide a syzygy analysis that makes explicit the obstacles.*

Assume (K1) holds. Write $A = \mathbb{C}[O_{\det_m}]$*,* $A_1 = W^*/I_1$ *(where* I_1 *is the degree-*1 *part of the ideal, which vanishes since* $O_{\det_m}$ *spans* W*). The multiplication map*

$$\phi_2 \colon \mathrm{Sym}^2(A_1) \to A_2$$

has kernel $K_2 := I(O_{\det_m})_2/(I(O_{\det_m})_1 \cdot A_1)$*, the space of quadratic relations. Under (K2),* $I(O_{\det_m}) = (K_2)$ *as an ideal.*

*For (K2) to hold, all syzygies among the generators must be expressible as linear combinations of degree-*2 *generators. Concretely, the first syzygy module* $\mathrm{Syz}_1(I(O_{\det_m}))$ *must be generated in degree* 3 *(one step above the generators). This is the defining property of a quadratic ideal being Koszul: the Koszul complex*

$$\cdots \to A(-3)^{b_2} \to A(-2)^{b_1} \to A \to \mathbb{C} \to 0$$

is exact, with all differentials linear.

Degree-*3 *syzygy obstruction. *For* $m = 2$*: the variety* $O_{\det_m}$ *inside* $W = \mathrm{Sym}_2(\mathbb{C}^{4*})$ *is related to the* 2×2 *determinantal locus; the ideal* $I(O_{\det_m})$ *is known to be generated by degree-*2 *polynomials (from the Plücker relations of* $\bigwedge^2 \mathbb{C}^4$*), and (K2) holds. For* $m = 3$*: the Cauchy decomposition gives* $A_1 \cong W^*$ *of dimension* 165*, and* $\mathrm{Sym}^2(A_1)$ *has dimension* $\binom{166}{2} = 13\,695$*. The target* A_2 *is the degree-*2 *part of the coordinate ring, whose dimension is not known explicitly but is bounded by* $\binom{167}{2}$*/(degree-2 relations)* $\approx 13\,861 - |K_2|$*. The* H*-equivariant decomposition of* K_2 *is controlled by the kernel of* $\mathrm{Sym}^2(W^*) \to \mathbb{C}[O_{\det_3}]_2$*; this kernel contains at minimum the components indexed by partitions* $(\lambda; \lambda)$ *with* $|\lambda| = 6$ *that* do not *appear in* $\mathbb{C}[O_{\det_3}]_2$*.*

By the analogous structure theory for classical determinantal varieties of matrices (not symmetric tensors), the first syzygies among degree-2 generators appear in degree 3, and higher syzygies persist through degree m (see Weyman [11], Chapter 6). For $m \geq 3$, the analogous structure for $I(O_{\det_m})$ (now an ideal in the symmetric algebra $\mathbb{C}[W]$) is expected to require generators in degrees ≥ 3, making (K2) false for $m \geq 3$.

More precisely: the orbit closure $O_{\det_m}$ is not *a determinantal variety in the classical sense (those are varieties of matrices of bounded rank), but rather the GL_{m^2}-orbit closure of a single polynomial in $\mathrm{Sym}_m(\mathbb{C}^{m^2*})$. The ideal $I(O_{\det_m})$ is generated by the GL_{m^2}-irreducible submodules of $\mathbb{C}[W]$ that vanish on $O_{\det_m}$. For $m = 3$, identifying these submodules is precisely the content of the Macaulay2 computation in Script 2 of Appendix B. Based on dimension arguments—the expected Hilbert function of $O_{\det_3}$ grows as $(d+1)^{\dim O_{\det_3}}/(\dim O_{\det_3})!$ to leading order in d—the ideal $I(O_{\det_3})$ is expected to contain degree-3 generators that do not follow from the degree-2 generators via linear combinations.*

Summary of status.

(i) (K2) is proved *for $m = 2$ (classical Plücker relations).*

(ii) (K2) is open *for $m = 3$; the Macaulay2 computation of $I(O_{\det_3})_2$ and $I(O_{\det_3})_3$ (Script 2, Appendix B) would decide it.*

(iii) (K2) is expected to fail *for $m \geq 4$ by analogy with classical resolutions of determinantal varieties [11].*

Conjecture 10.2 is offered in full awareness of this uncertainty: its content is that if *(K1)+(K2) hold, the Koszul property follows via the standard Priddy criterion [8]. An alternative approach for $m \geq 3$, not requiring (K2), would be to use the N-Koszul theory of Berger [2] for algebras with generators in degree 1 and relations in degree $N \geq 3$; this is not pursued here but represents a natural extension.*

10.2 Weight semigroup of $O_{\det_m}$

Conjecture 10.4 (Weight semigroup under H)**.** *The H-weight semigroup of $O_{\det_m}$ is*

$$\mathcal{X}_H(O_{\det_m}) = \{(\lambda; \lambda) \in \Lambda^+(H) : \ell(\lambda) \leq m,\ \lambda_1 \leq m\},$$

i.e., the set of H-dominant weights of the form $(\lambda; \lambda)$ where λ fits in an $m \times m$ box.

10.3 Newton–Okounkov bodies

Conjecture 10.5 (Newton–Okounkov volume separation)**.** *For all $n \geq 3$ (with m chosen so that* **VP** = **VNP** *requires $O_{\mathrm{perm}_n} \subseteq O_{\det_m}$):*

$$\mathrm{vol}\,\Delta(O_{\mathrm{perm}_n}) > \mathrm{vol}\,\Delta(O_{\det_m}).$$

This volume inequality would provide an asymptotic reformulation of the separation problem, independent of the algebraic milestones of Section 9. It is not implied by the results of the present paper.

11 Conclusion

What has been established

1. **Module–algebra gap** (Proposition 3.1). Graded G-algebras with identical isotypic multiplicity tables can have distinct SRMT values. The BIP theorem does not obstruct SRMT-based separation arguments for abstract graded algebras. The orbit-closure analogue is Open Problem (S4).

2. **Ray-Multiplicity-1 and Bilinear Rank-1 Rigidity of the determinant** (Theorems 4.4 and 5.4). $c_{k\lambda_{\det}}(\mathbb{C}[O_{\det_m}], k) = 1$ for all $k \geq 0$, proved directly from the character identity without any sphericity assumption. Consequently, $\mathrm{sr}^H_{O_{\det_m}}(a\lambda_{\det}, b\lambda_{\det}; (a+b)\lambda_{\det}) = 1$ for all $a + b \geq 1$, and the isotypic sub-algebra along the $\lambda_{\det}$-ray is isomorphic to $\mathbb{C}[t]$.

3. **Dimension obstruction for the permanent, and non-sphericity of its orbit** (Theorem 5.7 and Proposition 8.1). $\dim L^H(\mu_{\mathrm{perm}}) = \binom{2n-1}{n}^2 \gg 1$ prevents any analogue of Bilinear Rank-1 Rigidity. The orbit $G \cdot \mathrm{perm}_n$ is not spherical for all $n \geq 2$ (not merely $n \geq 3$ as in earlier versions); the explicit dimension comparison is given in the proof of Proposition 8.1. The LR coefficient $N^{2\mu_{\mathrm{perm}}}_{\mu_{\mathrm{perm}},\mu_{\mathrm{perm}}} = 1$ (Pieri rule); SRMT flexibility is conditional on (S1).

4. **SRMT–syzygy correspondence** (Theorem 6.2). The rank-nullity exact sequence links SRMT values to isotypic syzygies and cosyzygies, connecting the SRMT framework to equivariant free resolutions [11].

Open problems

The separation programme is contingent on (S1)–(S3). Open Problem (S4) (orbit-level multiplicity–SRMT independence) is structurally important for the general programme but not required for the specific separation via (S1)–(S3). Natural further directions include: verifying Conjecture 4.6 for $m = 3$; deciding (K2) for $m = 3$ via Macaulay2 (Script 2); computing Newton–Okounkov volumes for $n = 3$; extending the SRMT framework to $n \geq 4$; and developing N-Koszul theory for $\mathbb{C}[O_{\det_m}]$ when (K2) fails.

A Notation Summary

Symbol	Meaning
$G = \mathrm{GL}_{m^2}$	Full linear group acting on W
$W = \mathrm{Sym}_m(\mathbb{C}^{m^2*})$	Space of degree-m forms in m^2 variables
$H = \mathrm{GL}_m \times \mathrm{GL}_m$	Subgroup via $(A, B) \cdot X = AXB^T$
$O_{\det_m} = \overline{G \cdot \det_m}$	G-orbit closure of $\det_m$ (Zariski)
$O_{\mathrm{perm}_n} = \overline{G \cdot \mathrm{perm}_n}$	G-orbit closure of perm_n (Zariski)
Λ^+	Dominant integral weights of G (or H)
$L(\lambda)$	Irreducible G-module of highest weight λ
$c_\lambda(V)$	Multiplicity $\dim \mathrm{Hom}_G(L(\lambda), V)$
$N^\nu_{\lambda\mu}$	Littlewood–Richardson coefficient
$\mathrm{sr}_A^{d_1,d_2}(\lambda, \mu; \nu)$	SRMT value at degree pair (d_1, d_2)
$\delta_A^{d_1,d_2}(\lambda, \mu; \nu)$	SRMT defect
$\mathcal{X}(O)$	Weight semigroup of O
$\lambda_{\det} = (\omega_m; \omega_m)$	H-highest weight of $\det_m$ (one-dimensional)
$\mu_{\mathrm{perm}} = (m\omega_1; m\omega_1)$	H-highest weight of perm_n (dimension $\binom{2m-1}{m}^2$)
$\widehat{\det}_m \in A_1$	Linear form on $O_{\det_m}$ of H-weight $\lambda_{\det}$
$\mathrm{Syz}^{d_1,d_2}(A)_{\lambda,\mu;\nu}$	Isotypic syzygy kernel
$\mathrm{CoSyz}^{d_1,d_2}(A)_{\lambda,\mu;\nu}$	Isotypic cosyzygy cokernel
$R_{\lambda_{\det}}(O_{\det_m})$	$\lambda_{\det}$-ray isotypic sub-algebra $\cong \mathbb{C}[t]$

B Macaulay2 Code

The following scripts are conceptual implementations for the $n = 3$ separation programme. Due to the computational complexity of direct Gröbner elimination (see Remark 9.2), the recommended execution strategy is the character-theoretic approach via `SchurRings`.

Script 1: Ring setup and polynomial declarations

```
-- srmt_n3.m2 Script 1: Setup for the n=3 SRMT computation
-- Recommended: use SchurRings for isotypic decomposition
-- Direct Groebner elimination is infeasible (90 variables)
needsPackage "SchurRings"
kk = QQ
R = kk[x11,x12,x13,x21,x22,x23,x31,x32,x33,
        MonomialOrder => GRevLex]
perm3 = x11*x22*x33 + x11*x23*x32
      + x12*x21*x33 + x12*x23*x31
      + x13*x21*x32 + x13*x22*x31
det3  = x11*x22*x33 - x11*x23*x32
      - x12*x21*x33 + x12*x23*x31
      + x13*x21*x32 - x13*x22*x31
-- H-weights (at highest-weight level, not torus character level):
-- det3:  H-weight ldet = (omega_3; omega_3)  [one-dimensional]
-- perm3: H-weight mperm = (3*omega_1; 3*omega_1) [dimension 100]
-- Torus weight of BOTH is identical: (1,1,1;1,1,1)
-- This is not the H-highest weight; see Remark 2.6
```

Script 2: Isotypic decomposition via SchurRings (feasible)

```
-- Strategy: equivariant Hilbert series + character decomposition
-- This avoids the infeasible parametric elimination in 90 variables
-- Step 1: Compute H-equivariant Hilbert series of C[O_perm3]
--         using the known character of perm3 in W
-- Step 2: Decompose in SchurRing(QQ, s, 3) x SchurRing(QQ, t, 3)
S = schurRing(QQ, s, 3)
T = schurRing(QQ, t, 3)
-- mperm corresponds to s_{3} * t_{3}  (i.e., s_{(3)} tensor t_{(3)})
-- ldet  corresponds to s_{1,1,1} * t_{1,1,1}
-- Open Problem (S1): compute c_mperm(C[O_perm3], d0) for d0=2,3,...
-- Expected: c_mperm >= 2 at some d0
-- Open Problem (S3): compute c_{2*mperm}(C[O_det3])
-- = c of s_{6}*t_{6} in C[O_det3]
```

Script 3: SRMT computation (depends on S1)

```
-- Assumes: basis {v_i} of (C[O_perm3]_{d0})_mperm found from S1
-- r = c_mperm(C[O_perm3], d0) >= 2
-- Construct r x r multiplication matrix M:
--   M_{i,j} = component of v_i * v_j in (C[O_perm3]_{2*d0})_{2*mperm}
-- Open Problem (S2): verify rank(M) >= 2
-- If (S1)+(S2)+(S3) all hold:
--   sr^{d0,d0}_{O_perm3}(mperm,mperm;2*mperm) >= 2
--   sr^{d0,d0}_{O_det3}(mperm,mperm;2*mperm)  <= 1
--   => O_perm3 not subset O_det3  (Corollary 7.3)
--   => VP != VNP
```

C Proof of SRMT Monotonicity

Theorem C.1 (SRMT Monotonicity at fixed degree pairs)**.** *Let $O_1 \subseteq O_2$ be G-orbit closures in W. For any fixed integers $d_1, d_2 \geq 0$:*

$$\mathrm{sr}_{O_1}^{d_1,d_2}(\lambda,\mu;\nu) \leq \mathrm{sr}_{O_2}^{d_1,d_2}(\lambda,\mu;\nu) \quad \textit{for all } \lambda,\mu,\nu \in \Lambda^+.$$

Proof. **Step 1 (The surjection).** The Zariski-closed inclusion $O_1 \subseteq O_2$ induces a surjection of graded G-algebras $\varphi\colon \mathbb{C}[O_2] \twoheadrightarrow \mathbb{C}[O_1]$, $f \mapsto f|_{O_1}$, which respects the grading and commutes with the G-action.

Step 2 (Surjectivity on isotypic components). For each fixed degree d and each λ, the component map $(\varphi_d)_\lambda\colon (\mathbb{C}[O_2]_d)_\lambda \twoheadrightarrow (\mathbb{C}[O_1]_d)_\lambda$ is surjective (since φ_d is surjective and G-equivariance forces isotypic components to map surjectively). The degree pair (d_1, d_2) is the same for both orbit closures.

Step 3 (Commutativity with multiplication). Since φ is an algebra homomorphism, for the fixed degree pair (d_1, d_2):

$$\gamma \circ m_\nu^{O_2} = m_\nu^{O_1} \circ (\alpha \otimes \beta),$$

where $\alpha = (\varphi_{d_1})_\lambda$, $\beta = (\varphi_{d_2})_\mu$, $\gamma = (\varphi_{d_1+d_2})_\nu$ are all surjective.

Step 4 (Rank inequality). Since $\alpha \otimes \beta$ is surjective, $\operatorname{rank}(m_\nu^{O_1}) = \operatorname{rank}(m_\nu^{O_1} \circ (\alpha \otimes \beta))$. By Step 3, $m_\nu^{O_1} \circ (\alpha \otimes \beta) = \gamma \circ m_\nu^{O_2}$, so

$$\operatorname{rank}(m_\nu^{O_1}) = \operatorname{rank}(\gamma \circ m_\nu^{O_2}) \leq \operatorname{rank}(m_\nu^{O_2}),$$

where the inequality holds because post-composing with γ cannot increase rank. $\square$

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